University of the Philippines Diliman
College of Science
National Institute of Physics

Physics 180 THV-1 Problem Set 4-5

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IV - BS Physics

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Problem 4.1

There are situations when a radioactive decay results in a product nucleus that is also radioactive. We can have a decay chain 1 to 2 to 3 and so on, with the original nucleus (Type 1) being the parent. Assume that we begin with N_0 atoms of the parent (i.e., $N_1(0) = N_0$) and no decay products are originally present (i.e., $N_i(0) = 0, i = 1, 2, 3, \ldots$). Refer to the decay constants as $\lambda_1, \lambda_2, \lambda_3, \ldots$

We nowconsider the decay 1 to 2 to 3, where nuclei of type 3 are stable.

(a) Set up and solve the differential equation for $N_3(t)$.

(b) Evaluate $N_1(t) + N_2(t) + N_3(5)$ and interpret.

(c) Examin $N_1(t), N_2(t), N_3(t)$ at small t and interpret.

(d) Find the limits of $N_1(t), N_2(t), N_3(t)$ as t approaches infinity and interpret.

Solution

Problem 5.1

The energy of an alpha particle in terms of the Q-value was obtained by assuming a nonrelativistic form for the kinetic energy. Using ***relativistic expressions*** for p and T, **derive the relativistic version** of the equation we have seen in the lecture

$$T'_\alpha = \frac{M_D}{M_\alpha + M_D} Q = \frac{1}{1 + \frac{M_\alpha}{M_D}} Q \tag{2.1}$$

and calculate the error made by neglecting these corrections.

Solution

References

Appendices

Code

No code was used for this problem set